



HERMESHA RAO N S

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Objective

Seasoned Railway Signaling and System Assurance professional with 26 years of end-to-end experience delivering EPC and Turnkey railway projects across mainline railways, metro systems, CBTC projects, High-Speed Rail (HSR), freight corridors, freight yards, and railway yard modernization domains. Expertise spans system architecture, design, integration, testing, validation, commissioning, and safety assurance of mission- and safety-critical railway subsystems including Electronic Interlocking (EI), KAVACH (TCAS), ATP/ETCS Level 2, CBTC, MSDAC, ABS/e2ABS, point machines, AFTC, axle counters, and track circuits.

Extensive experience in Indian Railway projects involving signaling systems, railway station and signaling buildings, MEP coordination, depot and yard infrastructure, freight terminals, HSR corridors, and integrated rail infrastructure works within complex EPC environments. Possess over 20 years of experience in System Engineering and more than 15 years in RAMS and Functional Safety, leading complete safety life-cycle activities including PHA, FHA, SHA, IHA, OSHA, HAZOP, FMECA, FTA, SIL allocation, SIL 2 & SIL 4 compliance demonstration, and Independent Safety Assurance in accordance with EN 50126, EN 50128, EN 50129, IEC standards, and Indian Railway specifications.

Proven capability in delivering complete EPC/Turnkey life-cycle execution from concept development, system requirement specification, interface management, detailed engineering, installation supervision, FAT/SAT, integrated testing, field trials, CRS support, safety case preparation, certification, and final commissioning, ensuring fail-safe architecture, operational reliability, regulatory compliance, and on-time project delivery.

Strong expertise in multidisciplinary coordination with OEMs, EPC contractors, consultants, Independent Safety Assessors (ISA), civil, track, MEP, Telecom, and OHE stakeholders for seamless execution of complex railway infrastructure and signaling projects. Seeking to contribute as a senior leader, system assurance specialist, or delivery partner in global EPC/Turnkey railway projects, driving safe, salable, future-ready, and compliant signaling and transportation solutions across international rail networks.

● Freelance / Independent RAMS & Safety Assurance Consultant (Rail Systems)

Client: Hafeet Rail Project, / Alpha Energy (Oman)

Duration: 20th Nov 2024 – 30th Mar 2026 (18 Months)

Project Scope: MEP Safety Verification & System Engineering – Packages E2301, E2302, E2303 (Chain age & Boundary A/B/C)

Experience

- 30-03-2026
IRSE Indian Selection Committee / Alpha Energy Oman
Head of RAMS and System Engineering and (I Eng Chartered Assessment)
 - Leading technical discussions with global OEM partners (Korea, Malaysia, Canada, UAE) covering Electronic Interlocking (EI), ABS/e2ABS, ATP/ETCS, RFID Readers, and RFID Tag architecture, ensuring alignment of system design, performance, and client functional requirements.
 - Provided technical leadership during bid and execution phases, ensuring compliance with RDSO, IRS,
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and CENELEC requirements and alignment with client.

Led system integration and validation strategy for multi-vendor environments, coordinating signalling, telecom, rolling stock, and field interfaces.

Successfully supported EPC and Turnkey railway signalling projects across the full lifecycle from concept, system architecture, and scope definition through design, integration, testing, validation, and commissioning.

- Preparing complete BOQ (Bill of Quantities) for signalling materials as per client specifications, including EI/ABS equipment, Point Machines, Track Circuits, cables, relays, and ATP/RFID components, ensuring accuracy, compliance, and alignment with approved system architecture.
- Defined end-to-end signalling scope for EPC tenders, including EI, ATP/KAVACH, ABS/e2ABS, interfaces, RAMS requirements, V&V strategy, and acceptance criteria. Led the conversion of relay room interlocking (RRI) System with integration of VDU based control and introduction of UFSBI.
- Actively supported RDSO Vendor Registration and Product Approval, including:
 - Technical document review and gap analysis.
 - Compliance mapping against RDSO/SPN and IRS specifications
 - Safety justification and SIL evidence review
 - Interface validation and application logic assessment
 - Support during RDSO inspections, queries, and clarifications.
 - Conducted RAM analysis and ensured compliance with international safety and EN Standards.
 - Contributed to safety case preparation demonstration SIL-4 compliance and system acceptance.

- KAVACH Architecture and System Validation RDSO(IRSE India Selection) 28-10-2024
Lead RAMS R&D Validation Project (I Eng Chartered Assessment)

- Prepared and executed system-level V&V strategy, including test scenarios for SPAD prevention, overspeed protection, movement authority supervision, and degraded-mode operation
- Led end-to-end validation of the KAVACH ATP (Automatic Train Protection) system, covering system architecture review, interface validation, safety function verification, and fail-safe behavior assessment across onboard, trackside, and telecom subsystems. ◦ Reviewing and validating ATP hardware architecture, including onboard controller modules, vital processors, communication interfaces, OBC–RBC links, RFID reader–tag processing, and integration with TIU, brake interface unit, and locomotive electronics.
- Ensuring ATP software and hardware design complies with safety-critical development standards, including EN 50126, EN 50128, EN 50129, IEC 61508, and RDSO ATP application logic requirements.
- Supported EPC tender activities for KAVACH deployment by defining technical scope, validation requirements, compliance matrices, and acceptance criteria aligned with RDSO specifications.
- Conducting interface validation between KAVACH ATP, RFID Tags, RFID Readers, Brake Interface Unit, Loco Pilot DMI, and onboard safety relays, ensuring deterministic and failsafe data exchange.
- Leading safety assessments for ATP, including PHA, SHA, FHA, FMECA, FTA, SIL allocation/demonstration, and validating compliance evidence for certification and deployment.
- Successfully enabled allocation of the KAVACH trial section, coordinating with Railway authorities, OEMs, and project stakeholders, ensuring technical compliance and operational readiness.
- Knowledge on Software Testing Standards IEC 29119.
- Functional Block and System Requirements Test Case Procedures Verification Report Preparation.
- KAVACH GP& GA Safety Case Preparation;
- KAVACH Sw Tools Verification Report & Sw Release Note Preparation, Preparation of
- RAMS plan, Safety Plan, analysis reports to meets.
- KAVACH project internal ISA and interface with Extranal.
- Control Table Verification and Preparation of Tag & TIN Interlocking SIP.

- Alstom Rail Transportation Pvt Ltd

19-09-2023 - 14-06-2024

Safety Assurance Engineer Project

- Evaluation of Analysis PHA,SHA,IHA,OSHA, HZ LOG,SRAC,FMECA,FTA Safety Reports with HIRA & Risk Matrix Assessment.
 - Completed training on DOORS,AEC,Clear Quest, SAP and 8D problem solving skills. Urbils U400
 - ATC and ATS , Data Communication System Detailed company training completed.
 - Safety Report Analysis Agra and Kanpur Metro U200 Project. SSHA & IXL
 - Safety assessment DSP project Sydney.
- Indian Railway (Southern Railway) 23-08-2012 - 07-08-2023
Senior Signal / Junior Engineer (Signal) RAMS Engineer
 - Electronic Interlocking Specification Application Layout and Safety Case Preparation. Plan &
 - execute systems testing schedules with client within specified timelines.
 - Review and Validation of Design and Drawings for all Signaling Systems & Telecom systems.
 - TPWS/ ECTS L1 25 km Installation & Commissioning and Maintenance Over the RAMS 10 years experience FMEA with Microchip Reliability MTBF & MTTR evaluation report analysis. Procurement
 - of cost and Materials Room installation and cable Trenching with OFC cable and location boxes I EI System Installation and VDU with Embedded systems.
 - Prepare:- control plan, Track bonding plan,SIP Signal Interlocking plan, Route control table, location
 - boxes design plan and FAT& SAT Testing.
 - RAMS with isograph reliability workbench fault tree analysis System and Sub- System and On board track side 8 years.
 - On board track side Equipment Electrical Point Machine shall be installed and Ground fittings on site safety Correspondence Test and RAMS isograph reliability workbench Fault Tree with MTBF & MTTR evaluation reports.
 - Out Door Signaling:- Procurement of the aspect fixing assembly, Signal post mounting and aspect control Relay installation and RAMS fault tree Automatic Block section MTBF & MTTR .
 - Power Equipment:- Procurement of the Signal power supply arrangement Stabilizer, Chargers,Inverter Installation and isograph reliability workbench fault tree RAMS 15 years.
- Southern Railway 2004 - 2012
Technician Grade 1/ RAM systems
 - Procurement of the ELB and Machnical level Crossing Gate Installation and Ground fittings connection;
 - Electrical Point Machine shall be installed and Ground fittings on site safety
 - Correspondence Test and calibration;
 - RRI British Relay Interlocking system Installation and Wiring assemble Installation with RAMS isograph reliability workbench failsafe Experience 10 years.
- Southern Railway 03/2000 - 11/2003
Technician Grade II
 - On site switches, Boxes,SPD,Circuit breakers and Rotary switch Installation and equipment testing RAMS isograph reliability workbench and Water Fabrication data (High temperature Operating)5 years.
 - Bridge Channel Sleeper project Organization.
- Southern Railway 18-01-1999 - 12-03-2000
Assistant Coaching and EMU Care
 - Modern ICF Coach and EMU Trailer Coaches Maintenance;
 - Electrical Traction WAP4 Gear Box replacement and brake system Installation;
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Electrical Traction Loco Pantagraph Installation ;
TPWS Equipment assemble and installation EMU DV Coach

Engineering Council (UK) Specification IRSE IEng Incorporated Engineer No.725974 (Railway Signaling)	2025
• IRSE Indian Selection Committee Member Signal Expert Selection Committee Member	2025
• INTELLEX (IRSE 9001:2015) Advance Certificate Course in Railway RAMS, System Engineering and Assurance The course is accredited by the institution of Railway Signal Engineer-IRSE (UK).	2025
• IRSE (UK) 2024-2026 IRSE Membership AMIRSE (Associate Member) (00082370)	
• Rail Tel 1.3.180 IRSE Principal Tester License Assessment professional ongoing	2025
• Shivaji University (Kolhapur) Bachelor of Engineering (ECE) First Class	2004-2008
• Indian Institute of Chemical Engineer CRC CHENNAI (Anna University) Process Functional Safety Management First	2025
• Central Bharat Sevak Samaj (Govt of India) Post Diploma in Fire & Industrial Safety Management First Class	2023-2024
• Railway Signal and Telecom Training Center Signal and Telecom Training (CENELEC EN-50126,50129/CBTC-KAVACHI/TCAS)) A+	2000-2001 (2020)
• NEBOSH IGS Enrollment Assessment	2026
• MSME-TECHNOGY DEVELOPMENT CENTER(Govt of India) Chennai CERTIFIED LEAN SIX SIGMA BLACK BELT Complete	2023
• IPCS GLOBAL TECHNOLOGY Certified Embedded System Engineer & IoT	2023

Skills

- Engineering Drawings, CENELEC Standard Application (EN 50126, 50128,50716 & IEC 61511, 61508)FAT Testing and SAT Testing,Serial Standards and RAMS Isograph reliability workbench fault tree +.
- RAMS(Reliability, Availability, Maintainability & Safety) LOPA & Risk graph methods and Safety Integrity Level (SIL) Identify ,EUCCS, ETCS & Signal for high speed TWC, ATC, ATP,ATO,ATS Commands,On-board equipment,DOOR Supervision and Control system,Signal System.

- Fundamental of C Programming, Embedded C, Microcontroller and Embedded Systems, Arduino, PIC, ARM78051, STM32, Python Programming, Raspberry pi 3, Sensors interfacing, Camera interfacing, Communication Protocol (UART, USART, I2C, SPI, CAN), simultaneously Testing (MPLAB, PROTEUS DESIGN).
- AUTO CAD Drawings, Microstation Projects, C Programming, C++, Python Programming, Core Java, MS Office (Word, Excel, Power Point, HTML).
- Certified Lean SIX SIGMA BLACK BELT:-Responsible process improvement project implementation and cost reduction. Responsible for teaching Green belt project on process improvement projects. Monitor cost reduction Objectives and present results to upper management support. Manufacturing processes from design phase to start production. Managed simplified failure strategy and inventory reduction. Ensured adherence strict project timelines. FMEA Analysis and Problem solving methods.
- Functional Safety Process Management
- Reliability workbench Tools

Projects

- KAVACH Project R&D Product Developed
KAVACH project Develop Product GP with HW Design and Application Development of KAVACH GA & SA Prepare with SW & System Functions. and developed simulation for a Modbus TCP network and its security system.
- Electronic Interlocking System/ Microprocessor base Interlocking (2018-2022)
Electronic Interlocking with Windows/ Linux Embedded Standards (RDSO/ SPN/144) FAT Testing and SAT Testing (CENELEC Standards Application EN 50126, 50128, 50129) with SIL-4 safety.
- AFTC and MSDAC Projects 2014-2022 AFTC and MSDAC (Siemens) programming and Site Installation, Quad Cable laying and TPWS (Thales) Commission, ETCS System testing and CBTC Vital functions, TWC. IPS and HHSDAC 2022
- IPS and HHSDAC Axle Counters (Siemens/GG) programming and wiring assemble Installation Modern Signalling Training completed KAVACHI, TCAS, Metro ATS, ATC, ATP On-board, ETCS/ERTMS L2&L3.

Project Scope: MEP Safety Verification & System Engineering – Packages E2301, E2302, E2303 (Chainage & Boundary A/B/C) :- **Key Responsibilities & Deliverables:**

- Performed system-level safety verification of MEP (Mechanical, Electrical, Plumbing) systems during concept and early design stages of railway EPC packages
- Defined and validated System Boundaries, Interfaces, and Package Demarcations (Chainage-based segmentation A/B/C)
- Led development of Survey Baseline Documentation:
 - Route & Civil (LC, TVU, ROB/RUB, alignment, track structure)
 - Engineering / Structural layouts verification
 - Drainage & Plumbing risk mapping (flood, oil/chemical, fire, toxic gas pathways)
- Prepared and maintained Concept Hazard Log & Threat Log, including identification of interface risks between MEP and railway operations
- Developed RAMS & Security Management Plan (Concept Phase) aligned with EN 50126 lifecycle
- Established Requirements Traceability Matrix (RTM) linking system requirements to verification evidence
- Defined Verification & Validation (V&V) Strategy for MEP and interface systems
- Prepared Security Verification & Validation Plan focusing on infrastructure and operational safety risks
- Supported ISA coordination and safety compliance demonstration for early project approval stages

- KAVACH R&D Project on the time frame completed the plan for technical documentation and submitted the ISA clearance with Comments resolving.
- Successfully delivered multiple Electronic Interlocking (EI) Turnkey projects for Southern Railway, covering end-to-end lifecycle from system design through commissioning.
- Procurement of the project cost and System Modifications Circuit within the project time completion and handed over to Operating Department (2005-2012)
- Led and executed FAT (Factory Acceptance Testing) and SAT (Site Acceptance Testing), ensuring full compliance with RDSO specifications and project requirements.
- Coordinated with OEMs, site engineers, and Railway authorities to resolve non-conformities and achieve first-time-right commissioning.
- Electronic Interlocking System/Solid State Interlocking/ Microprocessor Interlocking System DUAL VDU modification and Installation with FMEA & FAT(2021-2022)